

EDUGENIE

6.1 Tailored Learning Pathways

Personalization and adaptive learning in EDUGENIE

Introduction

EDUGENIE can personalize an interactive lesson so that the same academic concept is presented through contexts that match a learner's interests, pace, preferred difficulty, and interaction style. The goal is not to change the learning objective, but to change the route through which the student reaches it.

Interest-Based Context Adaptation

- Projectile motion can be framed as a sports challenge, a space mission, a vehicle simulation, or a construction task.
- The mathematical and physical principles remain consistent while examples, visuals, goals, and narrative context change.
- Students can select a theme or the system can suggest one based on prior activity.

Personalization Dimensions

- Learning speed: provide more time, repetition, hints, or faster progression.
- Difficulty level: adjust variables, complexity, number of constraints, and challenge depth.
- Topic interests: connect lessons with sports, space, engineering, nature, games, or other student-selected themes.
- Preferred interaction: support text prompts, voice, buttons, drag-and-drop, or direct manipulation.
- Amount of guidance: move from guided examples to open-ended exploration.

Educational Benefits

- Can increase relevance and engagement by connecting abstract concepts to familiar contexts.
- Can support differentiated instruction within the same classroom.
- Can provide practice opportunities without forcing every student through exactly the same sequence.
- Can help teachers identify which types of support or examples are useful to individual learners.



Student data → adaptation → learning experience → feedback

Comparison / Learning Table

Dimension	Example adaptation	Learning purpose
Interest	Sports / space / vehicles	Same concept, more relevant context
Pace	Extra practice or accelerated tasks	Match learner progression
Difficulty	Basic → multi-variable challenge	Gradual mastery
Interaction	Voice / mouse / keyboard	Fit interaction preference
Guidance	Hints → independent mode	Build learner autonomy

Sample Coding Concept

```
# Simplified personalization logic
interest = "space" difficulty = 2 guidance = "medium"
if interest == "space": scenario = "planetary_mission"
elif interest == "sports": scenario = "basketball_shot"
else: scenario = "generic_projectile"
if difficulty >= 3: guidance = "low"
print(scenario, guidance)
```

Extra Project Content

- Classroom activity: Ask students to choose one context for the same physics objective, then compare whether the underlying concept changed.
- Assessment idea: Keep the learning objective constant while changing the generated scenario; assess concept mastery rather than story preference.
- Teacher dashboard opportunity: show progress, repeated errors, preferred contexts, and support levels without replacing teacher judgment.
- Future extension: use learner-controlled personalization so students can change context and difficulty rather than relying only on automatic decisions.